

Phase 6

Pull Request Submission Report

JGraphT Pull Request #1355

Refactor Boykov-Kolmogorov orphan adoption

GitHub account: Avraaaa

Fork: github.com/Avraaaa/jgrapht

Upstream: github.com/jgrapht/jgrapht

Verified: June 25, 2026

1. Phase 6 Requirement

The assignment requires:

- A fork of the original repository.
- A feature or refactoring branch.
- The improvement applied and pushed to that branch.
- A pull request submitted to the original repository.
- A clear pull request title.
- A description explaining the addressed issue, why the change improves the system, the refactoring approach, and assurance that functionality remains unchanged.

All required submission steps have been completed.

2. Repository and Pull Request Information

Item	Evidence
GitHub account	https://github.com/Avraaaa
Original repository	https://github.com/jgrapht/jgrapht
Forked repository	https://github.com/Avraaaa/jgrapht
Fork relationship	GitHub identifies Avraaaa/jgrapht as a fork of jgrapht/jgrapht
Refactoring branch	refactor-boykov-kolmogorov-adopt
Target branch	jgrapht:master
Pull request	https://github.com/jgrapht/jgrapht/pull/1355
Pull request number	#1355
Pull request title	Refactor Boykov-Kolmogorov orphan adoption
Commit	6183e894718be159d82a2182a72ca3bfb08b2a90
Files changed	1 production file
Diff size	152 additions and 105 deletions

The local Git configuration also confirms:

```
origin https://github.com/Avraaaa/jgrapht.git
upstream https://github.com/jgrapht/jgrapht
```

The local branch tracks the pushed fork branch:

```
refactor-boykov-kolmogorov-adopt
-> origin/refactor-boykov-kolmogorov-adopt
```

3. Pull Request Status

GitHub was checked on June 25, 2026.

Status item	Result
State	Open
Draft	No
Merge status	Not merged
Mergeability	Clean
Commits	1

Status item	Result
Changed files	1
Reviews	None submitted yet
Discussion comments	None after the opening description

The pull request was created on June 18, 2026 at 19:45:47 UTC. An open or unmerged state does not mean Phase 6 is incomplete: the assignment requires submission of a pull request, not acceptance by the upstream maintainers.

4. Verification Against the Instructor's PR Requirements

Instructor requirement	Evidence in PR #1355	Result
Clear title describing the improvement	Refactor Boykov-Kolmogorov orphan adoption	Satisfied
Explain what issue was addressed	Identifies cognitive complexity 112, Brain Method, and repeated source/sink adoption structure	Satisfied
Explain why the change improves the system	States that the large adoption logic is split into smaller private steps while retaining explicit directional cases	Satisfied
Summarize the refactoring approach	Lists extracted orphan selection, directional orphan handling, and directional parent search	Satisfied
Assure unchanged functionality	States that public API and algorithmic behavior are unchanged and identifies preserved invariants	Satisfied
Mention validation	Lists focused core tests, complete core tests, and aggregate Javadocs	Satisfied

The submitted description explicitly records that the refactor preserves:

- Orphan ordering.
- Source-tree and sink-tree edge orientation.
- Timestamp updates.
- Debug output.
- Public API behavior.
- Algorithmic behavior.

5. Continuous Integration Evidence

GitHub Actions ran the JGraphT pull request build on all configured operating systems:

Check	Result
Build (Windows)	Success
Build (Ubuntu)	Success
Build (macOS)	Success

All three checks completed successfully on June 18, 2026. This upstream CI evidence complements the local Phase 4 validation, where the original and refactored revisions both passed 31 focused tests and the refactored revision passed 102 maximum-flow regression tests.

6. Commit Evidence

Commit subject:

Refactor Boykov-Kolmogorov orphan adoption

Commit body:

Extract orphan selection from `adopt()`.

Extract source-tree and sink-tree orphan handling.

Extract source-tree and sink-tree parent search.

Preserve orphan ordering, edge orientation, timestamp updates, and debug output.

The commit is focused on one production file:

`jgrapht-core/src/main/java/org/jgrapht/alg/flow/BoykovKolmogorovMFIImpl.java`

7. Visual Evidence

7.1 Pull Request Submission

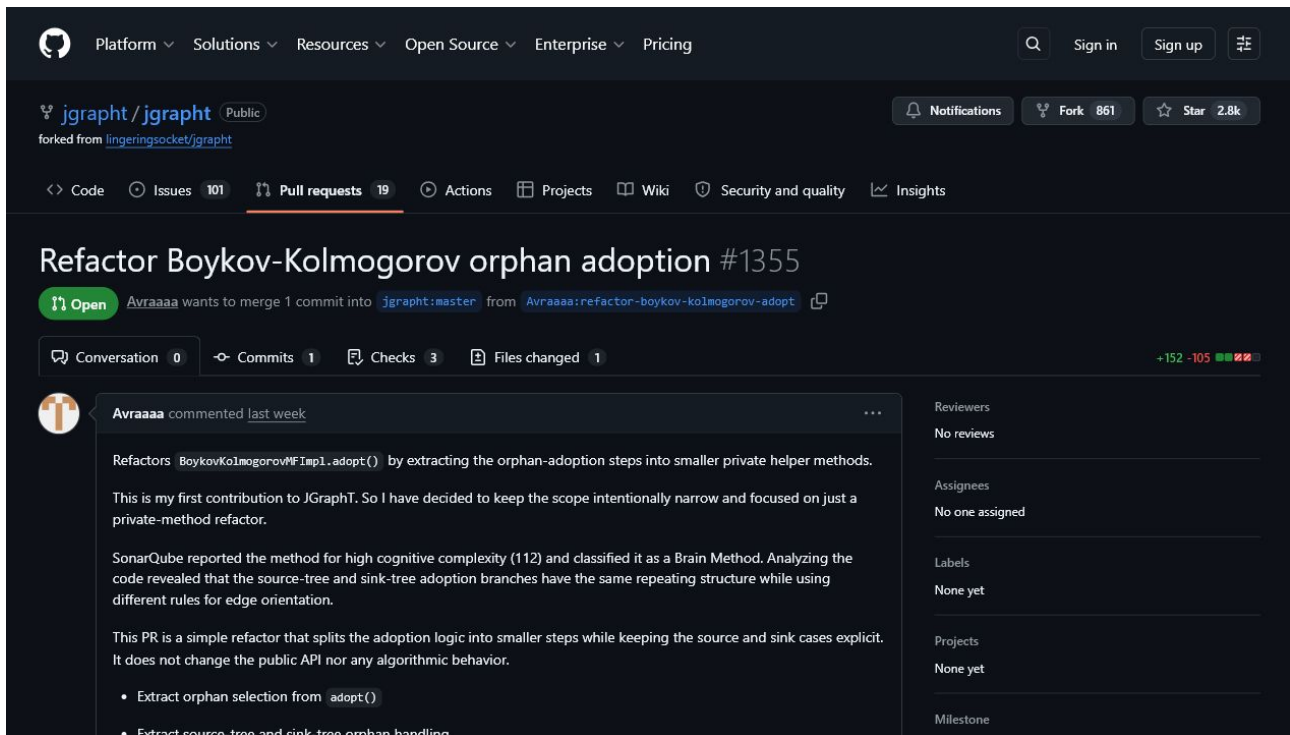


Figure 1. Open upstream pull request showing title, branches, state, commit count, changed file count, and opening description.

The screenshot `pull_request_1355.png` shows:

- PR number #1355.
- The clear refactoring title.
- Open status.
- Author Avraaaa.
- Source branch `Avraaaa:refactor-boykov-kolmogorov-adopt`.
- Target branch `jgrapht:master`.
- One commit and one changed file.
- The opening explanation of the code smell and refactoring.

7.2 CI Checks

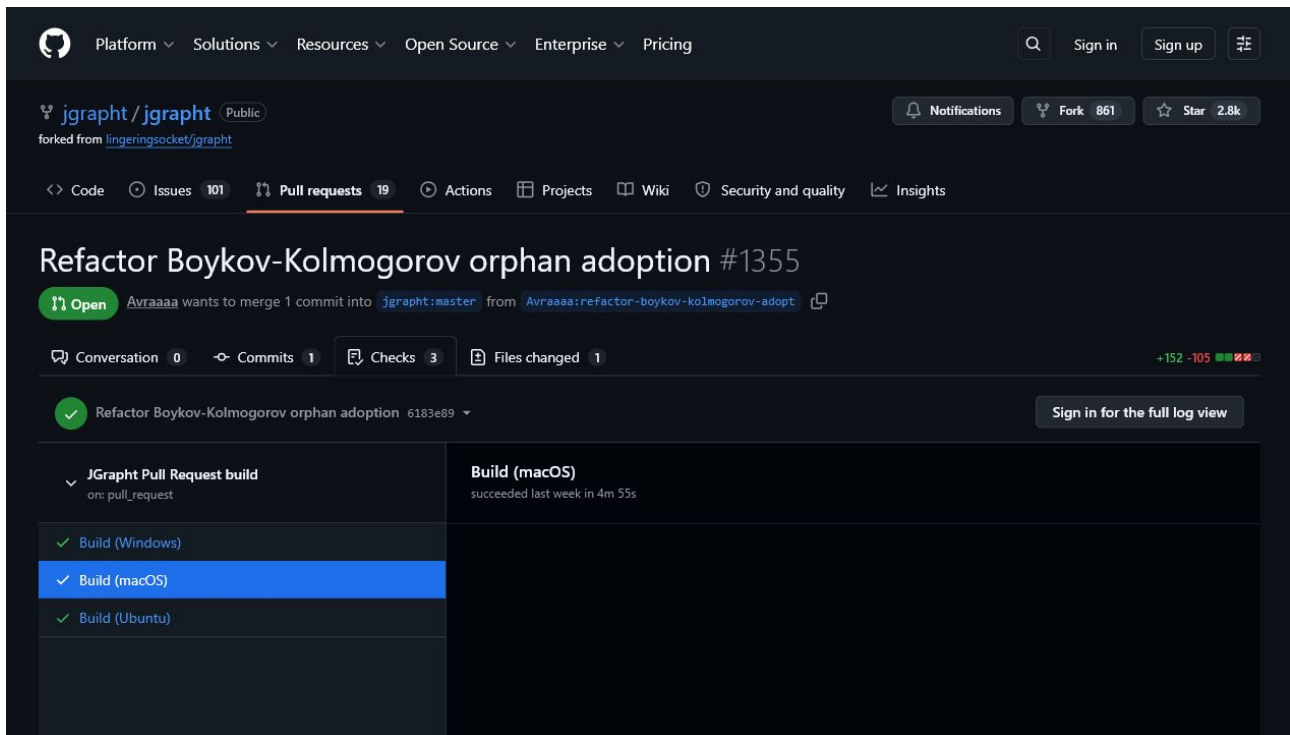


Figure 2. GitHub Actions checks showing successful Windows, macOS, and Ubuntu builds.

The screenshot `pull_request_1355_checks.png` shows successful Windows, macOS, and Ubuntu builds.

8. Phase 6 Completion Checklist

- [Completed] Forked the original repository.
- [Completed] Created a dedicated refactoring branch.
- [Completed] Applied and committed the improvement.
- [Completed] Pushed the branch to the fork.
- [Completed] Submitted a pull request to the original repository.
- [Completed] Used a clear title.
- [Completed] Explained the addressed maintainability issue.
- [Completed] Explained why the refactor improves maintainability.
- [Completed] Summarized the refactoring approach.
- [Completed] Assured that public behavior remains unchanged.
- [Completed] Included test commands in the PR description.
- [Completed] Passed all upstream GitHub Actions checks.

9. Conclusion

Phase 6 is complete. The change was submitted from the fork and dedicated refactoring branch to the original JGraphT repository as pull request [jgrapht/jgrapht#1355](#).

The live pull request satisfies the instructor's required title and description content. As of June 25, 2026, it is open, non-draft, cleanly mergeable, and has successful CI results on Windows, Ubuntu, and macOS.